Tension superficial

Dos fluidos → superficie plana

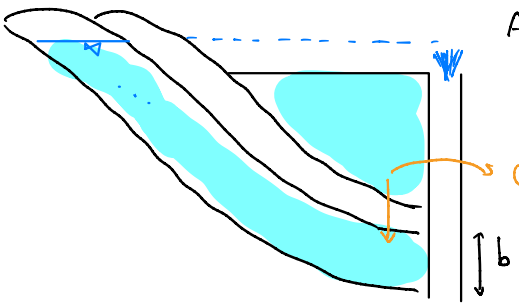
Capilaridad

Capacidad de subir o bajar en contra de la gravedad

Como calculo las conductividades equivalentes

Acuifero confinado

Recarga



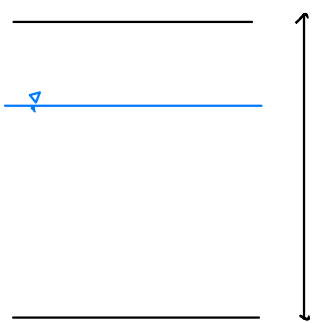
$$Ss = \frac{V_w}{V_{acui} \times \Delta h} = L^{-1}$$

S: almacenamiento

$S = Ss \times b = 1$

$Ss \sim 10^{-3} - 10^{-6}$

Acuifero Valle del Cauca



$$S = S_y + S_s \times b$$

Drenaje — Descompresion

$$q = k \frac{\Delta h}{\Delta l} \rightarrow q = -k \nabla h$$

K_{xx} K_{yy} K_{zz}

kh

Dupuit - Forcheimer

Repaso:
- Funciones de Bessel
- Transformada de Laplace

$$\int_A q \cdot n \cdot dA = \int_V \nabla \cdot q \cdot dv = \text{Flujo Neto}$$

Sy Acuifero libre $h = z$ (canal), Drenaje — Ecuacion de flujo

Ss Acuifero confinado $h = z + \frac{p}{\gamma}$ (tuberia), Potencia — $\nabla \cdot (K \nabla h) = \frac{S \partial h}{\partial t}$

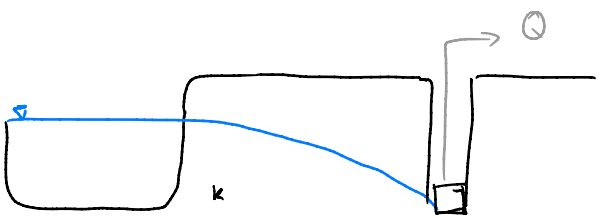
Flujo neto — $\frac{\partial h}{\partial t} \Delta s$

$S = S_y + S_s \times b$ $\quad S = \frac{V_w}{A_{ac} \cdot \Delta h}$ $\quad$ OUTPUT - INPUT

$\boxed{T \nabla^2 h = S \frac{\partial h}{\partial t}}$ $\quad D_H = \frac{S}{T}$ difusividad hidraulica

Acuifero confinado

$$\frac{\partial^2 h}{\partial x^2} + \frac{\partial^2 h}{\partial y^2} = 0$$



$q = K \frac{dh}{dx}$

Caudal unitario

$q' = -K \frac{d2h}{2dh}$

$q \cdot \frac{K d^2 h^2}{dx^2} - W = 0 = h^2 - h_1^2 - \frac{(h_1^2 - h_2^2)}{l} x + \frac{W}{K}(l - x)x$ $\quad \frac{q'}{K} dx = -h dh$

$d = \frac{l}{2} - \frac{K}{W} \frac{(h_1^2 - h_2^2)}{2l}$ $\quad h_{max} = \sqrt{h_1^2 - \frac{(h_1^2 - h_2^2)}{l} d + \frac{W}{K}(l-d) d}$

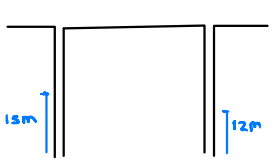
Ejm: $k = 10\,m/d$ $\quad h_1 = 15m$ $\quad -W = 60mm/a$ $\quad q_1'$

$S_y = 0.23$ $\quad h_2 = 18m$ $\quad +W = 20mm/a$ $\quad q_2'$

$b = 30m$ $\quad l = 200m$ $\quad d$

h_{max}

$h^2 - h_1^2 - \frac{(h_1^2 - h_2^2)}{l} x + \frac{W}{K}(l - x)x$, sin recarga, $W = 0$



$q' = \frac{K}{2l}(h_1^2 - h_2^2) - W\left(\frac{l}{2} - x\right)$

Ejercicio 6 costal:

Acuifero tipo dupuit, agua se infiltra

W: Recarga (mm/año), la dificultad de atravesar K (conductividad hidraulica)

- K grande (facil, arena), K pequeño (dificil, arcilla)

- DUPUIT: En vez de resolver el flujo en 3 dimensiones, asumamos que el agua se mueve horizontalmente Porosidad efectiva: 0.35

- El agua salada pesa mas que la dulce

Cerca de la costa, el agua salada de mar se mete por debajo del acuifero, formando una cuña

X_T: Punta de la cuna salina

Zona interior: $x = 0$ a $x = x_T$

Zona int. salina: $x = x_T$ a $x = L$

Profundidad de la cuna: $z = \frac{\rho_f}{\rho_s - \rho_f} \times h_{freatico} = 40 \times h$

Cambio climatico: . Sube el nivel del mar Δs, $\Delta L_SLR = -\Delta S / \tan\theta$ ↳ Retroceso tierra

. Erosion costera: ΔL_CE (Perdida extra de longitud)

Efecto total: $\Delta L = \Delta L_SLR + \Delta L_CE = \frac{-\Delta S}{\tan\theta} - \Delta L_CE$

Nuevo largo del acuifero: $L_b = L_0 + \Delta L$ Nuevo nivel del mar: $S_t = S_0 + \Delta s$

Símbolo	Significado	Valor
W	recarga por lluvia	500 mm/año
L_0	largo inicial del acuífero	1000 m
K	conductividad hidráulica	1×10^{-4} m/s
n_e	porosidad efectiva	0.35
ρf	densidad agua dulce	1.000
ρs	densidad agua salada	1.025
$\Delta\rho$	$\rho s - \rho f$	0.025
S_0	profundidad del fondo del acuífero bajo el nivel del mar	30 m

Nuevos, del Ejercicio 5 (el escenario de cambio climático):

Símbolo	Significado	Valor
tanθ	pendiente de la playa	0.02 (2%)
ΔS	aumento del nivel del mar	0.6 m
ΔL_CE	erosión costera	3 m

W: 500 mm/año = 0.5m/año

- Cuanto retrocede la costa:

$$\Delta L: \frac{-\Delta s}{\tan\theta} - \Delta L_CE = -(0.6/0.02) - 3$$

Preguntas:

1. Cuanto avanza la cuña de agua salada tierra adentro?

$$x_T = \sqrt{L^2 - \frac{(\rho_s \cdot \Delta\rho \cdot k)}{(\rho_f^2 \cdot w)} \times s^2}$$

$$C = \frac{(\rho_s \cdot \Delta\rho \cdot k)}{(\rho_f^2 \cdot w)} = \frac{1.025 \times 0.025 \times 1\times10^{-9}}{1^2 \times 1.58\times10^{-8}}$$

C / J

Antes del cambio climatico

$$x_T_0 = \sqrt{1000^2 - 161.8 \times 30^2} \quad L_0 = 1000,\ s_0 = 30$$

$$= 929.4\,m$$

Despues del cambio Climatico: ($L_t = 967$, $s_t = 30.6$)

$$x_T_f = \sqrt{967^2 - 161.8 \times 30.6} = 885.2\,m$$

$$\Delta x_T = x_T_f - x_T_0 = -39.1m$$

2. Cuanto sube el nivel freatico

Zona interior, Zona intrusion

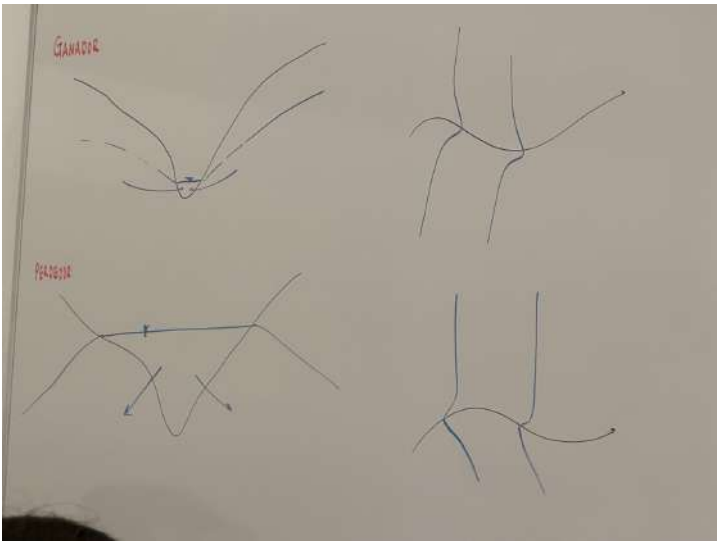
1. El mar subio
2. El acuifero es mas corto ahora

Evaluada antes y despues

$$H_Iz(x) = \sqrt{\frac{W}{k} \times (L^2 - x^2) + \frac{\rho_s}{\rho_f} \times s^2}$$

$$I_Iz\ (x=10m) = H_Iz,t(10) - H_Iz,0(10) = 0.41m$$

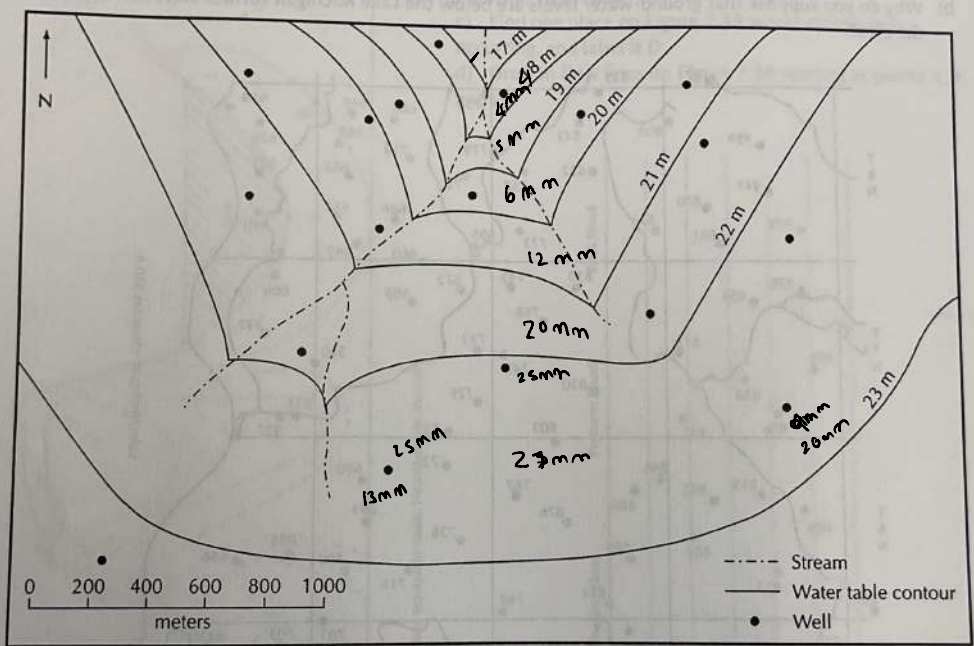
Taller en clase



CORRECCION ATMOSFERICA

CORRECCION POR MAREA

1. The black dots on Figure 3.31 represent the location of water-table wells. Based on the location of the ground-water contour lines, estimate the elevation of the water table in each of the wells to 0.1 m. Calculate the greatest and least values of the hydraulic gradient found in Figure 3.31.



▲ FIGURE 3.31
Water-table map with the hydraulic gradient changing across the site. Where the contour lines are close together, the gradient is steeper than where they are farther apart. Note that the contour lines form the shape of a V where they cross a stream.

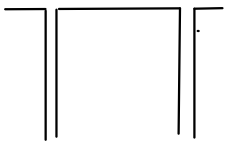
▲ FIGURE 7.39
Diagram for Problem 3. The dashed lines are equipotentials. Heads are in meters.

3. Answer the following questions based on Figure 7.39.

a) Fill in the heads at the locations labeled on the diagram.
b) Find one place on Figure 7.39 where recharge is occurring, and label it R.
c) Find one place on Figure 7.39 where discharge is occurring, and label it D.
d) Draw in flow lines on Figure 7.39 starting at points X, Y, and Z.

Location	Elevation Head	Pressure Head	Total Head
A			
B			
C			
D			
E			
F			

$$\frac{1}{r}\frac{\partial A}{dr} + \frac{\partial^2 A}{\partial r^2} = \frac{S}{T}\frac{\partial A}{\partial t}$$



Formula:

$$T = \frac{2.303\,Q}{2\pi(S_1 - S_2)} \times \log_{10}\left(\frac{r_2}{r_1}\right)$$

r_1: 0.8 r_2: 30

S_1: 2.236 S_2: 1.088

Q: 788

$$T = \frac{2.303\,Q}{2\pi(2.236 - 1.088)} \times \log_{10} \times \left(\frac{30}{0.8}\right) = 0.503 \times Q =$$

Transmisividad y almacenamiento

VERDE

W: 1 Q: 500 gal/min $T = \frac{Q \times W(v)}{4\pi \times s} = \frac{Q \times 1}{4\pi \times 0.56} = 0.142Q$

u: 0.1 96250 $ft^3/día$

S: 0.56 ft

r^2/t: 2.8×10^7 $ft^2/día$ $S = \frac{4Tu}{r^2/t} = \frac{4 \times T \times 0.1}{2.8 \times 10^7} = 1.43 \times 10^{-8} T$

240 Minutos 0.16

$$: 2\sqrt{\frac{Tt}{S}}$$

Eim:

$T = 100 m^2/d$

$S = 1 \times 10^{-4}$

$Q = 20 l/s$

$Q_2 = 50 l/s$

$d = 200 m$

$s \quad r_1 = r_2 = 50 m$

$t = 12 h$

$$s(r,t) = \frac{Q}{4\pi T} \ln\left(\frac{2.25 r^2 S}{4Tt}\right)$$

1) · Unidades

$t = 12h = 0.5d$

$Q_1 = 1728 m^3/d$

$Q_2 = 4320 m^3/d$

$$s = \ln \frac{2.25 \times 100 \times 0.5}{2500 \times 10^{-4}} = 6109$$

$4\pi T = 1256.6 m^2/d$

Descenso cada pozo:

$$S1) \frac{1728}{1256.6} \times 6109 = 8.4 m$$

$$S2) \frac{4320}{1256.6} \times 6109 = 21 m$$

3 efectos:
- Otro pozo
- Carga a nivel constante
- barrera impermeable

Superposicion:

$S_1 + S_2 = 29.4 m$

$$R: 1.5\sqrt{\frac{Tt}{S}}$$